

ML2 2.2

Complex Machine Learning Model

Keras and RNN

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RNN #1

Epoch: 30

Batch size: 32

Hidden layers: 32

Verbose: 1

Activation function: sigmoid

Loss: 0.3668 → 0.0893

Accuracy: 0.8255 → 0.9624

Note: There is great evidence of if loss improvement and accuracy improvement!

```
--- Starting Model Training (Many-to-Many) ---
Epoch 1/30
574/574 ----- 3s 3ms/step - accuracy: 0.8255 - loss: 0.3668 - val_accuracy: 0.9090 - val_loss: 0.2072
Epoch 2/30
574/574 ----- 1s 2ms/step - accuracy: 0.9018 - loss: 0.2201 - val_accuracy: 0.9176 - val_loss: 0.1849
Epoch 3/30
574/574 ----- 1s 2ms/step - accuracy: 0.9110 - loss: 0.2002 - val_accuracy: 0.9230 - val_loss: 0.1714
Epoch 4/30
574/574 ----- 1s 2ms/step - accuracy: 0.9165 - loss: 0.1880 - val_accuracy: 0.9284 - val_loss: 0.1614
Epoch 5/30
574/574 ----- 1s 2ms/step - accuracy: 0.9216 - loss: 0.1776 - val_accuracy: 0.9337 - val_loss: 0.1534
Epoch 6/30
574/574 ----- 1s 2ms/step - accuracy: 0.9250 - loss: 0.1702 - val_accuracy: 0.9360 - val_loss: 0.1464
Epoch 7/30
574/574 ----- 1s 3ms/step - accuracy: 0.9295 - loss: 0.1627 - val_accuracy: 0.9410 - val_loss: 0.1381
Epoch 8/30
574/574 ----- 1s 2ms/step - accuracy: 0.9311 - loss: 0.1563 - val_accuracy: 0.9444 - val_loss: 0.1312
Epoch 9/30
574/574 ----- 1s 2ms/step - accuracy: 0.9349 - loss: 0.1506 - val_accuracy: 0.9468 - val_loss: 0.1262
Epoch 10/30
574/574 ----- 1s 2ms/step - accuracy: 0.9381 - loss: 0.1429 - val_accuracy: 0.9489 - val_loss: 0.1191
Epoch 11/30
574/574 ----- 1s 2ms/step - accuracy: 0.9389 - loss: 0.1426 - val_accuracy: 0.9535 - val_loss: 0.1120
Epoch 12/30
574/574 ----- 1s 2ms/step - accuracy: 0.9435 - loss: 0.1320 - val_accuracy: 0.9547 - val_loss: 0.1081
Epoch 13/30
574/574 ----- 1s 2ms/step - accuracy: 0.9450 - loss: 0.1281 - val_accuracy: 0.9580 - val_loss: 0.1034
Epoch 14/30
574/574 ----- 1s 2ms/step - accuracy: 0.9478 - loss: 0.1225 - val_accuracy: 0.9610 - val_loss: 0.0979
Epoch 15/30
574/574 ----- 1s 3ms/step - accuracy: 0.9489 - loss: 0.1203 - val_accuracy: 0.9628 - val_loss: 0.0932
Epoch 16/30
574/574 ----- 1s 2ms/step - accuracy: 0.9509 - loss: 0.1161 - val_accuracy: 0.9643 - val_loss: 0.0894
Epoch 17/30
574/574 ----- 1s 2ms/step - accuracy: 0.9517 - loss: 0.1146 - val_accuracy: 0.9658 - val_loss: 0.0871
Epoch 18/30
574/574 ----- 1s 2ms/step - accuracy: 0.9537 - loss: 0.1101 - val_accuracy: 0.9684 - val_loss: 0.0815
Epoch 19/30
574/574 ----- 1s 2ms/step - accuracy: 0.9543 - loss: 0.1089 - val_accuracy: 0.9682 - val_loss: 0.0810
Epoch 20/30
574/574 ----- 1s 2ms/step - accuracy: 0.9554 - loss: 0.1060 - val_accuracy: 0.9710 - val_loss: 0.0765
Epoch 21/30
574/574 ----- 1s 2ms/step - accuracy: 0.9557 - loss: 0.1052 - val_accuracy: 0.9709 - val_loss: 0.0784
Epoch 22/30
574/574 ----- 1s 2ms/step - accuracy: 0.9574 - loss: 0.1033 - val_accuracy: 0.9721 - val_loss: 0.0740
Epoch 23/30
574/574 ----- 1s 2ms/step - accuracy: 0.9580 - loss: 0.1008 - val_accuracy: 0.9735 - val_loss: 0.0700
Epoch 24/30
574/574 ----- 1s 2ms/step - accuracy: 0.9596 - loss: 0.0964 - val_accuracy: 0.9745 - val_loss: 0.0672
Epoch 25/30
574/574 ----- 1s 2ms/step - accuracy: 0.9599 - loss: 0.0950 - val_accuracy: 0.9743 - val_loss: 0.0679
Epoch 26/30
574/574 ----- 1s 2ms/step - accuracy: 0.9600 - loss: 0.0951 - val_accuracy: 0.9763 - val_loss: 0.0650
Epoch 27/30
574/574 ----- 1s 2ms/step - accuracy: 0.9607 - loss: 0.0934 - val_accuracy: 0.9762 - val_loss: 0.0625
Epoch 28/30
574/574 ----- 1s 2ms/step - accuracy: 0.9621 - loss: 0.0899 - val_accuracy: 0.9776 - val_loss: 0.0604
Epoch 29/30
574/574 ----- 1s 2ms/step - accuracy: 0.9631 - loss: 0.0887 - val_accuracy: 0.9762 - val_loss: 0.0615
Epoch 30/30
574/574 ----- 1s 2ms/step - accuracy: 0.9624 - loss: 0.0893 - val_accuracy: 0.9787 - val_loss: 0.0585
```

RNN #2

Epoch: 30

Batch size: 40

Hidden layers: 38

Verbose: 2

Activation function: softmax

Loss: 12.4965 → 12.4695

Accuracy: 0.2161 → 0.2161

Note: I assume the softmax prevented the improvement of the loss and accuracy.

```
459/459 - 2s - 5ms/step - accuracy: 0.2161 - loss: 12.4965 - val_accuracy: 0.2105 - val_loss: 12.5872
Epoch 2/30
459/459 - 1s - 2ms/step - accuracy: 0.2161 - loss: 12.4965 - val_accuracy: 0.2105 - val_loss: 12.5872
Epoch 3/30
459/459 - 1s - 2ms/step - accuracy: 0.2161 - loss: 12.4965 - val_accuracy: 0.2105 - val_loss: 12.5872
Epoch 4/30
459/459 - 1s - 2ms/step - accuracy: 0.2161 - loss: 12.4965 - val_accuracy: 0.2105 - val_loss: 12.5872
Epoch 5/30
459/459 - 1s - 2ms/step - accuracy: 0.2161 - loss: 12.4965 - val_accuracy: 0.2105 - val_loss: 12.5872
Epoch 6/30
459/459 - 1s - 2ms/step - accuracy: 0.2161 - loss: 12.4965 - val_accuracy: 0.2105 - val_loss: 12.5872
Epoch 7/30
459/459 - 1s - 2ms/step - accuracy: 0.2161 - loss: 12.4965 - val_accuracy: 0.2105 - val_loss: 12.5872
Epoch 8/30
459/459 - 1s - 2ms/step - accuracy: 0.2161 - loss: 12.4965 - val_accuracy: 0.2105 - val_loss: 12.5872
Epoch 9/30
459/459 - 1s - 2ms/step - accuracy: 0.2161 - loss: 12.4965 - val_accuracy: 0.2105 - val_loss: 12.5872
Epoch 10/30
459/459 - 1s - 2ms/step - accuracy: 0.2161 - loss: 12.4965 - val_accuracy: 0.2105 - val_loss: 12.5872
Epoch 11/30
459/459 - 1s - 2ms/step - accuracy: 0.2161 - loss: 12.4965 - val_accuracy: 0.2105 - val_loss: 12.5872
Epoch 12/30
459/459 - 1s - 2ms/step - accuracy: 0.2161 - loss: 12.4965 - val_accuracy: 0.2105 - val_loss: 12.5872
Epoch 13/30
459/459 - 1s - 2ms/step - accuracy: 0.2161 - loss: 12.4965 - val_accuracy: 0.2105 - val_loss: 12.5872
Epoch 14/30
459/459 - 1s - 2ms/step - accuracy: 0.2161 - loss: 12.4965 - val_accuracy: 0.2105 - val_loss: 12.5872
Epoch 15/30
459/459 - 1s - 2ms/step - accuracy: 0.2161 - loss: 12.4965 - val_accuracy: 0.2105 - val_loss: 12.5872
Epoch 16/30
459/459 - 1s - 2ms/step - accuracy: 0.2161 - loss: 12.4965 - val_accuracy: 0.2105 - val_loss: 12.5872
Epoch 17/30
459/459 - 1s - 2ms/step - accuracy: 0.2161 - loss: 12.4965 - val_accuracy: 0.2105 - val_loss: 12.5872
Epoch 18/30
459/459 - 1s - 2ms/step - accuracy: 0.2161 - loss: 12.4965 - val_accuracy: 0.2105 - val_loss: 12.5872
Epoch 19/30
459/459 - 1s - 2ms/step - accuracy: 0.2161 - loss: 12.4965 - val_accuracy: 0.2105 - val_loss: 12.5872
Epoch 20/30
459/459 - 1s - 2ms/step - accuracy: 0.2161 - loss: 12.4965 - val_accuracy: 0.2105 - val_loss: 12.5872
Epoch 21/30
459/459 - 1s - 2ms/step - accuracy: 0.2161 - loss: 12.4965 - val_accuracy: 0.2105 - val_loss: 12.5872
Epoch 22/30
459/459 - 1s - 2ms/step - accuracy: 0.2161 - loss: 12.4965 - val_accuracy: 0.2105 - val_loss: 12.5872
Epoch 23/30
459/459 - 1s - 2ms/step - accuracy: 0.2161 - loss: 12.4965 - val_accuracy: 0.2105 - val_loss: 12.5872
Epoch 24/30
459/459 - 1s - 2ms/step - accuracy: 0.2161 - loss: 12.4965 - val_accuracy: 0.2105 - val_loss: 12.5872
Epoch 25/30
459/459 - 1s - 2ms/step - accuracy: 0.2161 - loss: 12.4965 - val_accuracy: 0.2105 - val_loss: 12.5872
Epoch 26/30
459/459 - 1s - 2ms/step - accuracy: 0.2161 - loss: 12.4965 - val_accuracy: 0.2105 - val_loss: 12.5872
Epoch 27/30
459/459 - 1s - 2ms/step - accuracy: 0.2161 - loss: 12.4965 - val_accuracy: 0.2105 - val_loss: 12.5872
Epoch 28/30
459/459 - 1s - 2ms/step - accuracy: 0.2161 - loss: 12.4965 - val_accuracy: 0.2105 - val_loss: 12.5872
Epoch 29/30
459/459 - 1s - 2ms/step - accuracy: 0.2161 - loss: 12.4965 - val_accuracy: 0.2105 - val_loss: 12.5872
Epoch 30/30
459/459 - 1s - 2ms/step - accuracy: 0.2161 - loss: 12.4965 - val_accuracy: 0.2105 - val_loss: 12.5872
```

RNN #3

Epoch: 30

Batch size: 50

Hidden layers: 44

Verbose: 1

Activation function: sigmoid

Loss: 0.3714 → 0.0789

Accuracy: 0.8193 → 0.9678

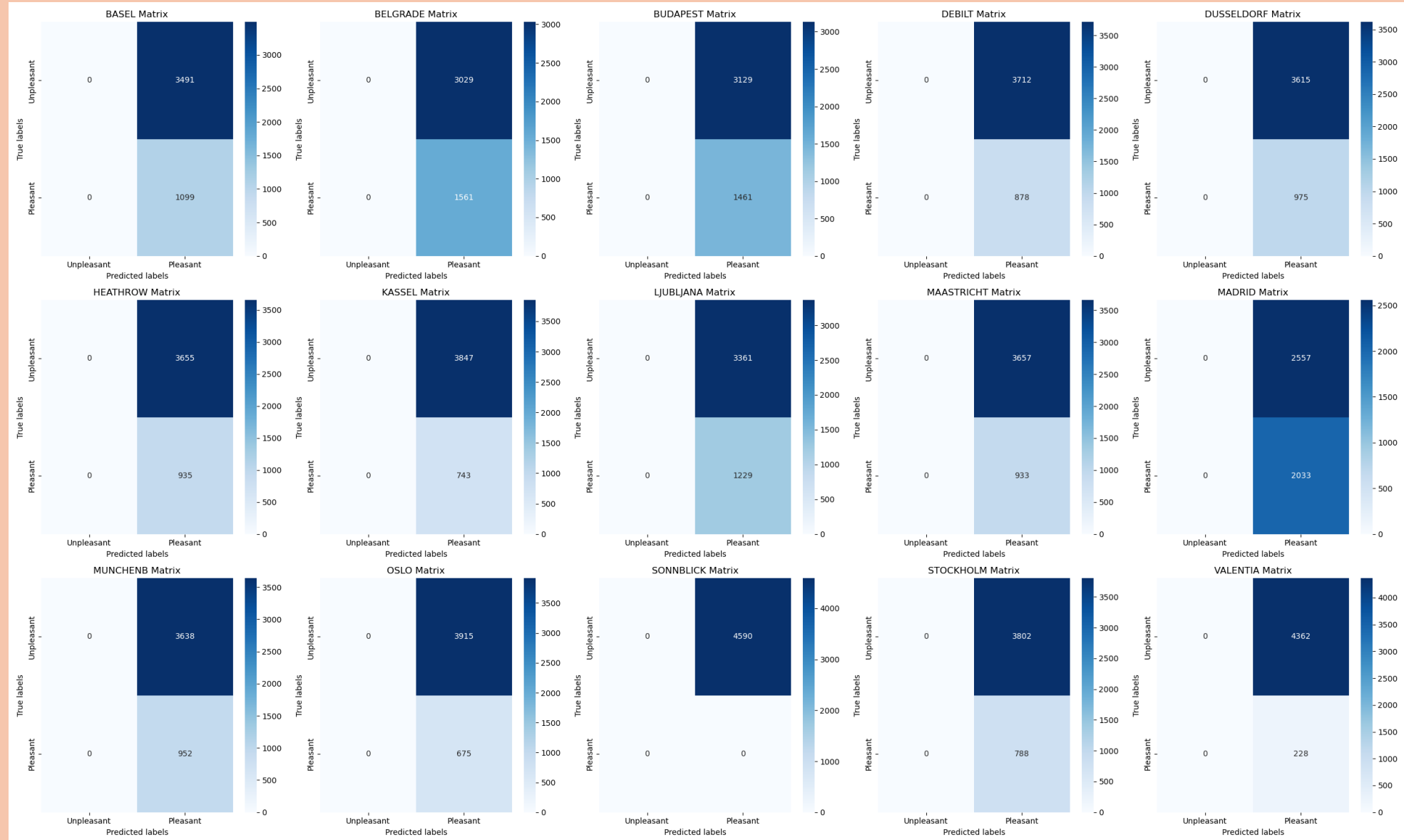
Note: This is a slight amount of
improvement from RNN #1.

```
--- Starting Model Training ---
Epoch 1/30
368/368 ----- 2s 3ms/step - accuracy: 0.8193 - loss: 0.3714 - val_accuracy: 0.9109 - val_loss: 0.2001
Epoch 2/30
368/368 ----- 1s 3ms/step - accuracy: 0.9060 - loss: 0.2127 - val_accuracy: 0.9193 - val_loss: 0.1821
Epoch 3/30
368/368 ----- 1s 3ms/step - accuracy: 0.9137 - loss: 0.1933 - val_accuracy: 0.9248 - val_loss: 0.1688
Epoch 4/30
368/368 ----- 1s 3ms/step - accuracy: 0.9179 - loss: 0.1868 - val_accuracy: 0.9299 - val_loss: 0.1593
Epoch 5/30
368/368 ----- 1s 3ms/step - accuracy: 0.9260 - loss: 0.1707 - val_accuracy: 0.9365 - val_loss: 0.1469
Epoch 6/30
368/368 ----- 1s 3ms/step - accuracy: 0.9303 - loss: 0.1618 - val_accuracy: 0.9413 - val_loss: 0.1387
Epoch 7/30
368/368 ----- 1s 3ms/step - accuracy: 0.9338 - loss: 0.1538 - val_accuracy: 0.9433 - val_loss: 0.1318
Epoch 8/30
368/368 ----- 1s 3ms/step - accuracy: 0.9369 - loss: 0.1495 - val_accuracy: 0.9483 - val_loss: 0.1246
Epoch 9/30
368/368 ----- 1s 3ms/step - accuracy: 0.9413 - loss: 0.1395 - val_accuracy: 0.9504 - val_loss: 0.1189
Epoch 10/30
368/368 ----- 1s 3ms/step - accuracy: 0.9422 - loss: 0.1368 - val_accuracy: 0.9527 - val_loss: 0.1143
Epoch 11/30
368/368 ----- 1s 3ms/step - accuracy: 0.9444 - loss: 0.1327 - val_accuracy: 0.9549 - val_loss: 0.1088
Epoch 12/30
368/368 ----- 1s 3ms/step - accuracy: 0.9481 - loss: 0.1253 - val_accuracy: 0.9589 - val_loss: 0.1037
Epoch 13/30
368/368 ----- 1s 3ms/step - accuracy: 0.9501 - loss: 0.1213 - val_accuracy: 0.9600 - val_loss: 0.0993
Epoch 14/30
368/368 ----- 1s 3ms/step - accuracy: 0.9512 - loss: 0.1177 - val_accuracy: 0.9610 - val_loss: 0.0957
Epoch 15/30
368/368 ----- 1s 3ms/step - accuracy: 0.9519 - loss: 0.1157 - val_accuracy: 0.9626 - val_loss: 0.0940
Epoch 16/30
368/368 ----- 1s 3ms/step - accuracy: 0.9541 - loss: 0.1110 - val_accuracy: 0.9667 - val_loss: 0.0874
Epoch 17/30
368/368 ----- 1s 3ms/step - accuracy: 0.9558 - loss: 0.1074 - val_accuracy: 0.9658 - val_loss: 0.0859
Epoch 18/30
368/368 ----- 1s 3ms/step - accuracy: 0.9570 - loss: 0.1045 - val_accuracy: 0.9674 - val_loss: 0.0832
Epoch 19/30
368/368 ----- 1s 3ms/step - accuracy: 0.9585 - loss: 0.1017 - val_accuracy: 0.9680 - val_loss: 0.0816
Epoch 20/30
368/368 ----- 1s 3ms/step - accuracy: 0.9586 - loss: 0.1006 - val_accuracy: 0.9692 - val_loss: 0.0788
Epoch 21/30
368/368 ----- 1s 3ms/step - accuracy: 0.9605 - loss: 0.0966 - val_accuracy: 0.9712 - val_loss: 0.0753
Epoch 22/30
368/368 ----- 1s 3ms/step - accuracy: 0.9612 - loss: 0.0949 - val_accuracy: 0.9715 - val_loss: 0.0748
Epoch 23/30
368/368 ----- 1s 3ms/step - accuracy: 0.9618 - loss: 0.0922 - val_accuracy: 0.9723 - val_loss: 0.0727
Epoch 24/30
368/368 ----- 1s 3ms/step - accuracy: 0.9632 - loss: 0.0893 - val_accuracy: 0.9731 - val_loss: 0.0708
Epoch 25/30
368/368 ----- 1s 3ms/step - accuracy: 0.9643 - loss: 0.0883 - val_accuracy: 0.9734 - val_loss: 0.0668
Epoch 26/30
368/368 ----- 1s 3ms/step - accuracy: 0.9647 - loss: 0.0862 - val_accuracy: 0.9743 - val_loss: 0.0666
Epoch 27/30
368/368 ----- 1s 3ms/step - accuracy: 0.9655 - loss: 0.0846 - val_accuracy: 0.9749 - val_loss: 0.0645
Epoch 28/30
368/368 ----- 1s 3ms/step - accuracy: 0.9659 - loss: 0.0825 - val_accuracy: 0.9749 - val_loss: 0.0622
Epoch 29/30
368/368 ----- 1s 3ms/step - accuracy: 0.9660 - loss: 0.0824 - val_accuracy: 0.9781 - val_loss: 0.0593
Epoch 30/30
368/368 ----- 1s 3ms/step - accuracy: 0.9678 - loss: 0.0789 - val_accuracy: 0.9761 - val_loss: 0.0642
```

Confusion Matrix

This multi-confusion matrix doesn't provide great insight. It reads as though there are low amount of pleasant weather days and more undefined days that are neither pleasant nor unpleasant.

However, the only other option for this is unpleasant since there are only 0s and 1s in the pleasant weather data set.



END